Accounting for Love of Variety in Consumer Demand Estimation

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August 23, 2026

1 Introduction

In the baseline logit model, a consumer is perceived as having a fixed preference for each product, and the utility from consuming a product is independent of past consumption. Further, it is assumed that a new good is always a utility boosting event, by the error term's property of being identically and independently distributed. However, in many markets, consumers may have a love of variety, or a desire to consume different products over time. This can be due to factors such as boredom, satiation, or a preference for novelty. In such cases, the utility from consuming a product may depend on past consumption, and the introduction of a new good may not always lead to an increase in utility.

I contribute to the love of variety literature by adding a supply side to the existing demand models. With the supply side, I am able to understand how firms will maximize profits using customers' desire for variety in product characteristics. Further, I develop the models by controlling for endogeneity in prices via a control function approach. This corrects potential bias in price coefficients that exist in previous work. I find that consumers do not gain utilty from variety. The estimation yields a statistically significant disutility for variety, implying that people prefer habit in an attribute like flavor. I also find that consumers would need to be paid between \$0.20 and \$0.30 to switch. This desire for habit creates a pathway for interesting firm-side experiments like targeted discounting or strategic product introduction.

The rest of the paper is organized as follows: section two discusses the related literature and section three describes the data used and some descriptive statistics. Section four showcases the model and environment and section five presents results. Section six goes into counterfactual analysis, while section 7 concludes.

2 Related Literature

The variety-seeking and state dependence literature has been primarily focused on understanding the dynamics of consumer preferences and demand over time. Beginning with **?**, who studied ... .From **?**, who developed an empirical structural model of state dependence in consumer choice, to **?**, who iterated the model to allow for consumer learning. **?** is the most closely related paper, as he develops a model of variety-seeking behavior which is both tractible and flexible, and uses Nielsen data to estimate the model. However, this paper only looks at demand, leaving me a niche to explore the interplay between variety-seeking and strategic advertising. **?** progresses Chintagunta's work, showing how consumers seek variety in quantities purchased. Using the soda market, Dube is able to model purchase occasions at a very granular level. **?** ... .

I also add to the demand estimation methodology literature. **?** introduces a control function methodology which accounts for price endogeneity, taking the residual from the first stage of an IV regression and adding it as an element of the utility function.

3 Data

3.1 Data

My data comes from Nielsen's HomeScan and MarketScan data. These two main datasets give me access to household purchase decisions as well as product characteristics. I am looking at the year 2014 in markets Atlanta, Chicago, San Diego, and Denver. Households are comprised of a single-person to avoid any contamination of shopping for multiple people like children or spouses, allowing me a cleaner sample of shopping trips. These trips are encoded with unique identifiers, thus allowing me to look at per-trip, and within-trip, decisions. Stores have to be valid Nielsen stores, meaning they appear in the reatil code data. This is necessary to observe prices the products were bought at. The product space I am looking at is yogurt. Yogurt is good for this sort of analysis due to its relative lack of storability as well as its variation in attributes like flavor, nutrient contents, and types like Greek or Skyr. I have limited yogurt sizes to cups. Cups of yogurt are between five and eight ounces and are single serving products which come in more varied flavors.

I classify flavors into three categories: berry, plain, and other. While these are not taking full advantage of the granularity in flavors available, the classifications capture the two largest categories, berry and plain. I define the outside option as a trip in which yogurt was not ever purchased. 1 shows that the outside option is taken 80% of the time each trip. My sample has 1,269 agents who are seen for an average of about 18 trips. Of those 1,269, 715 made a yogurt purchase in 2014. If an agent was a yogurt purchaser, they purchased about 2.3 cups of yogurt per-trip. The mean income of my sample was 18,000. The sample is predominately white, with 75% of agents being White, 20% being Black, and 5% being either Asian or Hispanic.

Conditional on purchasing yogurt, 57% of households switch flavors at least once during the observation period. Approximately 50% of switching households did so using a coupon during purchase. It is impossible to know what product the coupon was used on, given multiple units are purchased each period. Yogurt buying households, on average, buy the same flavor for ten consecutive trips and switch three times.

Looking at 1, we can see the breakdown of switching behavior by flavor. For those who switched from berry yogurt to plain yogurt, they will stay on plain for 4.4 trips on average. Those who switch to plain tend to stay on plain longer for either case, staying an average of four periods if they switch from either flavor to plain. For those switching to other, they stay on that other flavor for 3.2 trips no matter the origin flavor. For berry, it appears that households are less stuck to it or use it as their switching flavor. That is, they break up the monotony of their primary flavor, plain or other, with a berry yogurt. This is interesting, since plain is more expensive on average. This implies the household's taste for plain yogurt outweighs their disutility of cost.

4 Model

4.1 Environment

There are a continuum of $i \in \mathcal{I}$ consumers who exist for $\mathcal{T}$ periods. Within a given period, a household can go for a shopping trip $t \in \mathcal{T}$ which is comprised of choice occasions $o \in \mathcal{O}_t$. Each choice occasion allows for the purchase of a singular good.

4.2 Consumer's Problem

The consumer's problem is the following. In each period t, the consumer i chooses a product j from the menu of available products J_t. Each product $j \in J_t$ is a bundle of $k \in K$ characteristics such that $X_{jt}^k \in \mathcal{X}_{jt}$. The consumer's utility from choosing product j at time t is given by:

$$U_{ijt} = \beta_0 + \sum_1^k \beta_i^k X_{jt}^k + \gamma_i \Xi_{ijt} + \alpha_i p_{jt} + \sigma \hat{\eta}_{jt} + \varepsilon_{ijt}$$

subject to

$$\begin{aligned} \Xi_{ijt} &= |X_j t - X_{j,t-1}^*| \\ \gamma_i &\sim \mathcal{N}(0,1) \\ \varepsilon_{ijt} &\sim \text{T1EV}(0,1) \end{aligned}$$

In the utility function, there are the following components: β_0 is a constant, β_i^k is a consumer's utility from characteristic X_{jt}^k, γ_i is their preference for variety, Ξ_{ijt} is the difference between the current characteristic and the most recent choice, α_i is a hetergeneous price sensitivity to p_{jt} and ε_{ijt} is the Gumbel shock which allows for a closed form logit.

To correct for price endogeneity in the above, I have implemented a control function as in **?**. Here, I do a first stage instrumental variable estimation and then take residual from that estimation and insert it into the utility function. That is $\hat{\eta}$, which is recovered from:

$$p_{jt} = \bar{p}_{jt} + \tau_t + \eta_{jt}$$

Where $\bar{p}_{jt}$ is the mean price from the three markets the household is not shopping in and τ_t is a time indicator. The residual corrects for the bias present in pricing, allowing us a consistent estimate of pricing.

Choice probabilities take the following form:

$$P_{ijt} = \frac{\exp \beta_0 + \sum_1^k \beta_i^k X_{jt}^k + \gamma_i \Xi_{ijt} + \alpha_i p_{jt} + \sigma \hat{\eta}_{jt} + \varepsilon_{ijt}}{1 + \exp \sum_s \beta_0 + \sum_1^k \beta_i^k X_{st}^k + \gamma_i \Xi_{ist} + \alpha_i p_{st} + \sigma \hat{\eta}_{jt} + \varepsilon_{ist}}$$

Where we normalize the outside option, not purchasing yogurt, to be zero by construction.

5 Estimation

The estimation procedure uses standard maximum likelihood estimation, while implementing the **?** control function. This means that there is no random coefficient estimation at this point, allowing me to identify the parameters of interest from the data alone. This is accomplished by first making an index of prices, instrumental variable residuals, and previous choices for each household's trips for each flavor. Then, I plug these values into the objective function represented by: U

$$_{ijt} = \beta_0 + \beta_i^b X_{jt}^b + \beta_i^p X_{jt}^p + \gamma_i \Xi_{ijt} + \alpha_i p_{jt} + \sigma \hat{\eta}_{jt}$$

Where β_0 is a constant, β_i^b, β_i^p are choice specific constants for choosing berry or plain. Other flavors are left out to avoid multi-collinearity. $\gamma_i \Xi_{ijt}$ is the love of variety term, which is just capturing if you switched between periods. The price and control function term are unchanged from above. The probability of individual i choosing the observed product j in time t is given by:

$$\Pi_i (P_{ijt})^{y_{ijt}}$$

Where $y_{ijt} = 1$ when person i chosen j that period, and zero otherwise. The log-likelihood is then given by:

$$\mathrm{LL}(\theta) = \sum_i^I \sum_j y_{ijt} \ln P_{ijt}$$

I then maximize the total log-likelihood to recover the parameters at the optimal.

6 Results

Results are shown in table 2. Here, you can see mixed findings. Firstly, I find that γ_i is statistically significant and negative with a value of -1.23. This finding points to consumers being less willing to switch flavors between trips. This estimate is at odss with what the data implies: more than half of houses are switchers, conditional on yogurt purchase, and houses switch 3 times on average if they do switch. This discrepancy points to my sample potentially suffering from being too small. Further, I find a negative price coefficient α, but am unable to claim statistical significance. This is fixable with a better instrumental variable, something which is actively being considered. Similarly, the control function correction coefficient is not significant either. However, given σ is positive, we can infer that α is positively biased without this correction. With a better IV approach, the control function should also produce more impactful effects.

When looking at the characteristic coefficients, only berry, β_i^b, is statistically significant. This is, again, an issue with lack of observations. As table 1 shows, berry has over 2,000 observed times being chosen. When looking at plain's coefficient, β_i^p, I do not find significance. This makes total sense upon seeing it is only chosen 412 times. The constant is negative, which is to be expected with a high outside option uptake rate. It is close to fringe significance, but is not either.

Finally, I calculated a willingness-to-pay for all β coefficients as well as the variety coefficient γ. These values are found in table 3. For the intercept, consumers value the outside option approximately \$1.50 more than yogurt flavored "other". For berry, it is valued about \$1.81 more than other flavors, while plain is valued about \$0.77 more. More interestingly, I find that consumers would need to be heavily compensated to switch. They would need approximately \$2.20 in incentives to switch from their normal choice of flavor. Judging by the average pricing of yogurt, this implies needing about two to three cups of yogurt given to them for free to switch.

Overall, my findings suggest consumers prefer to stick on flavors, experiencing disutility from switching. While a surprising result given the data facts, it must be noted that consumers do not just buy flavor. They buy for other characteristics that are not accounted for at this point in the research process. It will be interesting to see if this changes once I create a more holistic bundle of characteristics for the consumer to have preferences over. Similarly, the willingness-to-pay for variety is -2.20, meaning the consumer would need to be compensated \$2.20 to switch flavors. This opens a door for interesting counterfactual exercises in which the supply side is aware of the consumers willingness-to-pay and preferences, and thus try to influence them to switch.

7 Counterfactuals

Now, suppose we introduce firm side dynamics. Here, the firm acts statically, taking the consumer's last observed action into account and deciding whether or not to offer a promotion to encourage switching. That is, in each period t, firm f maximizes profits according to the following problem:

$$\begin{aligned} V_{f,t} &= \max\{V_f^d, V_f^n\} \\ \text{subject to} & \\ V_f^d &= p'_j s_j(x_j, p'_j) - MC_j \\ V_f^n &= p_j s_j(x_j, p_j) - MC_j \\ p' &< p \end{aligned}$$

The firm has to decide whether to offer a discount, V_f^d, or sell at the normal price, V_f^n. Prices are set at p' when giving a discount, a level strictly lower than the baseline. I assume a flat marginal cost that does not change if a discount is offered. The firm has the following constraints:

$$\begin{gathered} V_f^d \geq V_f^n \\ p'_j s_j(p'_j, x_j) \geq p_j s_j(p_j, x_j) \end{gathered}$$

That is, the firm will advertise when the market share under the new price is larger than under the old price, offsetting the lowered price.

As an example, suppose the firm offers a discount of 20% on berry yogurt if you are a plain type. Thus, the consumer faces the following problem:

$$U_{ijt} = \beta_0 + \sum_{1}^{k} \beta_i^k X_{jt}^k + \gamma_i \log(1 + \Xi_{ijt}) + \alpha_i p'_{jt} + \sigma\hat{\eta}' + \xi_j + \varepsilon_{ijt}$$

subject to

$$\begin{aligned}
\Xi_{ijt} &= |X_j t - X^*_{j,t-1}| \\
p'_{jt} &= 0.8 * p_{jt} \\
\gamma_i &\sim \mathcal{N}(0,1) \\
\varepsilon_{ijt} &\sim \text{T1EV}(0,1)
\end{aligned}$$

This implies that the price of the consumer's non-current flavor will be lowered by 20%. For instance, if the consumer chooses plain, the firm will lower its berry yogurt price by 20%.

8 Conclusion

In closing, this paper develops a dynamic discrete choice model of consumer demand that incorporates a love of variety and strategic advertising by firms. The model is estimated using scanner data, and counterfactual analyses are conducted to evaluate the effects of different market structures on consumer welfare and market outcomes. The paper contributes to the literature on advertising and consumer behavior, by providing empirical evidence on the effects of strategic advertising in a dynamic demand setting. The findings of this paper have important implications for policymakers and firms, as they highlight the potential benefits and drawbacks of targeted advertising in markets when accounting for consumers' preference for variety.

9 Tables and Figures

Table 1: Summary Statistics for Household Panel, Switching Dynamics, and Pricing

Statistic / Variable	Value	Unit / Description
Panel & Household Demographics		
Total Households in Sample	1,269	Households
Yogurt Purchasing Households	715	Households (56%)
Mean Trips per Household	17.65	Trips / Household
Mean Purchases per Shopping Trip	5.23	Units / Trip
Mean Household Income	17.62	Income Bracket
Median Household Income	18.00	Income Bracket
Outside Option Rate	80%	Non-purchase trip share
Racial Composition of Households		
White (Code 1)	955	Households (75%)
Black (Code 2)	254	Households (20%)
Asian (Code 3)	24	Households (2%)
Hispanic (Code 4)	36	Households (3%)
Switching & Purchasing Dynamics		
Ever-Switcher Households	57%	Conditional on yogurt purchase
Coupon Purchase Rate (All Purchases)	46%	Share of yogurt trips
Switching Attributable to Coupon	49%	Share of switch events
Mean Times Switching per Household	3	Switches / Household
Mean Consecutive Buys per Flavor	10.16	Continuous purchase trips
Yogurt Pricing		
Overall Mean Price	\$0.84	Per unit price
Other Flavors Price	\$0.76	Per unit price
Berry Price	\$0.92	Per unit price
Plain Price	\$1.16	Per unit price
Yogurt Choices		
Outside Option Chosen	8,102	Times Outside Option Chosen
Berry Chosen	2,108	Times Berry Yogurt Chosen
Other Chosen	1,865	Times Other Yogurt Flavors Chosen
Plain Chosen	412	Times Plain Yogurt Chosen

Figure 1

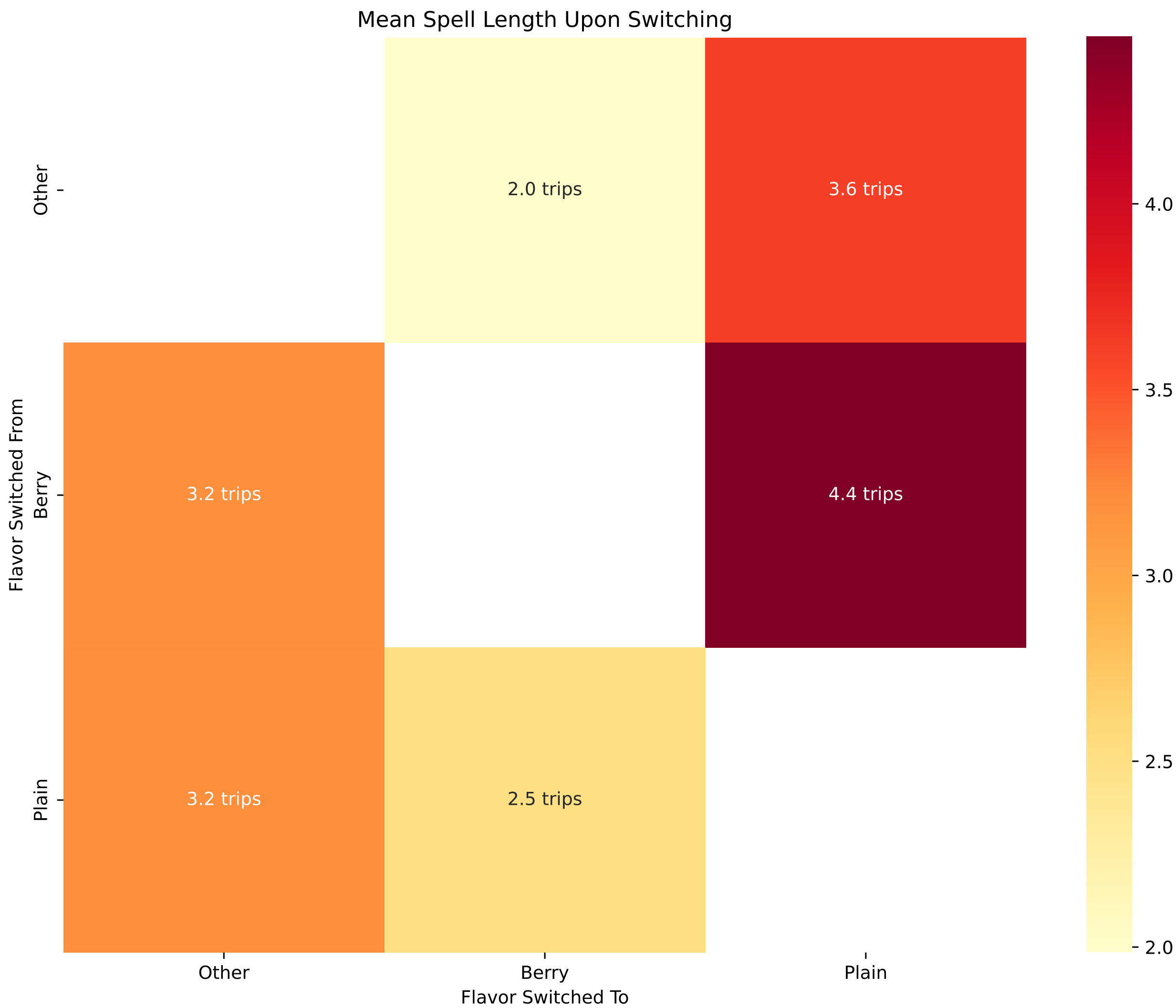
Mean Spell Length Upon Switching
2.0 trips
3.6 trips
3.2 trips
4.4 trips
3.2 trips
2.5 trips
Other
Berry
Plain
Flavor Switched From
Other
Berry
Plain
Flavor Switched To
4.0
3.5
3.0
2.5
2.0

Table 2: Structural Estimation Results

Parameter	Estimate	Std. Error	z-stat	p-value
β_0 (Intercept)	-0.8402	0.5736	-1.465	0.14
β_{ber} (Berry)	1.0115	0.2485	4.070	0.00^{***}
β_{pl} (Plain)	0.4322	0.7620	0.567	0.57
γ (Habit/Variety)	-1.2333	0.5372	-2.296	0.02^{**}
α (Price)	-0.5603	0.5762	-0.973	0.33
σ (Control Func)	0.3831	2.7661	0.139	0.89

$^{***}p < 0.001$, $^{**}p < 0.05$

Table 3: Implied Willingness-to-Pay (WTP) Estimates

Attribute / State	WTP ($)
Berry Flavor (β_{ber})	1.8053
Plain Flavor (β_{pl})	0.7714
Base Utility (β_0)	-1.5000
Habit/Variety (γ)	-2.2011

WTP calculated as $-$Parameter/α, where $\alpha = -0.5603$.